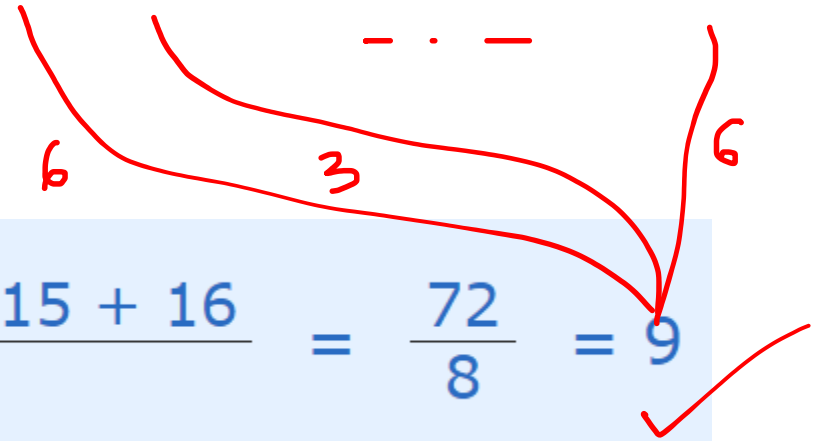
Fundamental of Statistics and its types

Mean Deviation

- How far, on average, all values are from the middle.
- In three steps:
- 1. Find the mean of all values
- 2. Find the **distance** of each value from that mean (subtract the mean from each value, ignore minus signs)
- 3. Then find the **mean of those distances**

Mean Deviation

- Example: the Mean Deviation of 3, 6, 6, 7, 8, 11, 15, 16
- Step 1: Find the **mean**:



$$\text{Mean} = \frac{3 + 6 + 6 + 7 + 8 + 11 + 15 + 16}{8} = \frac{72}{8} = 9$$

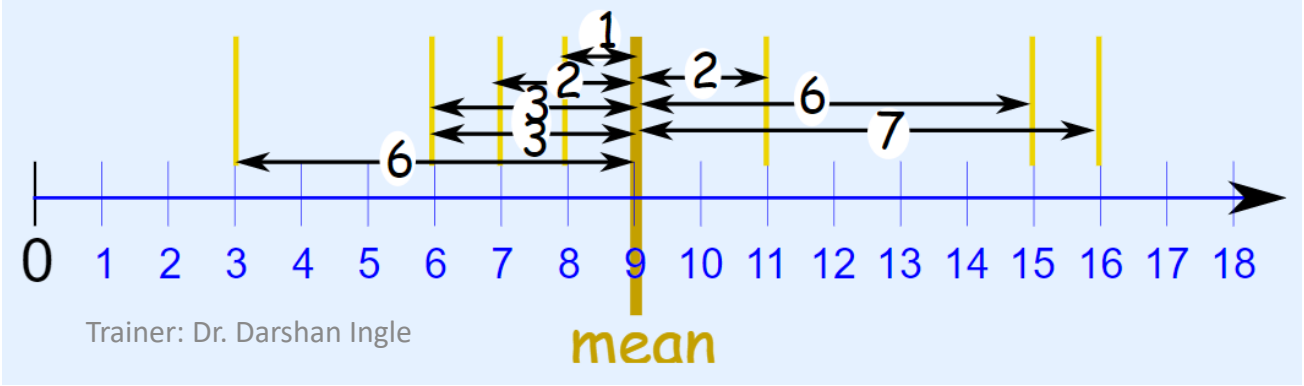


Mean Deviation

- Step 2: Find the **distance** of each value from that mean:

Value	Distance from 9
3	$\|3-9\|=6$
6	3
6	3
7	2
8	1
11	2
15	6
16	$\|16-9\|=7$

- Which looks like this:



Mean Deviation

- Step 3. Find the **mean of those distances**:

$$\text{Mean Deviation} = \frac{6 + 3 + 3 + 2 + 1 + 2 + 6 + 7}{8} = \frac{30}{8} = 3.75$$

- So, the **mean = 9**, and the **mean deviation = 3.75**
- It tells us how far, on average, all values are from the middle.
- In that example the values are, on average, 3.75 away from the middle.
- For **deviation** just think **distance**

Formula for Mean Deviation

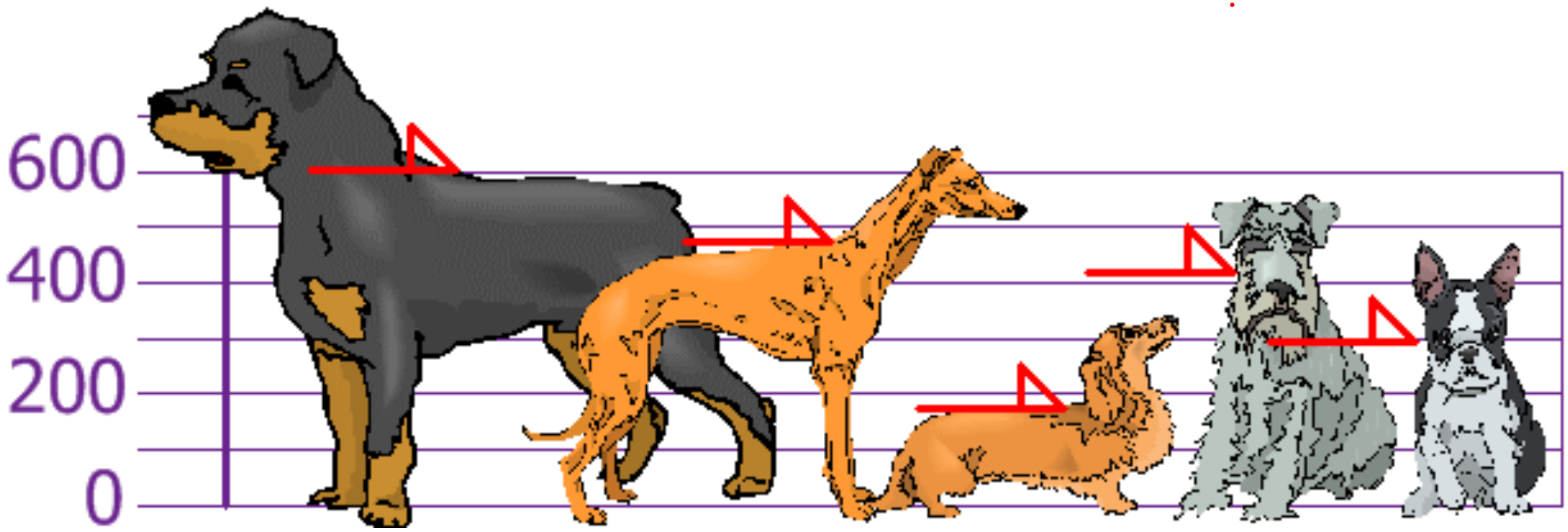
$$\text{Mean Deviation} = \frac{\Sigma|x - \mu|}{N}$$

- **Σ** is Sigma, which means to sum up
- || (the vertical bars) mean Absolute Value, basically to ignore minus signs
- **x** is each value (such as 3 or 16)
- **μ** is the mean (in our example **μ = 9**)
- **N** is the number of values (in our example **N = 8**)

- *Note: the mean deviation is sometimes called the Mean Absolute Deviation (MAD) because it is the mean of the absolute deviations.*

Mean Deviation

Example: You and your friends have just measured the heights of your dogs (in millimeters):



- The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.

Mean Deviation

Example: You and your friends have just measured the heights of your dogs (in millimeters):

- The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.
- Step 1: Find the **mean**:

$$\mu = \frac{600 + 470 + 170 + 430 + 300}{5} = \frac{1970}{5} = 394$$

Mean Deviation

- Step 2: Find the **Absolute Deviations**:

x	$\lvert x - \mu \rvert$
600	206
470	76
170	224
430	36
300	94
	$\Sigma \lvert x - \mu \rvert = 636$

Mean Deviation

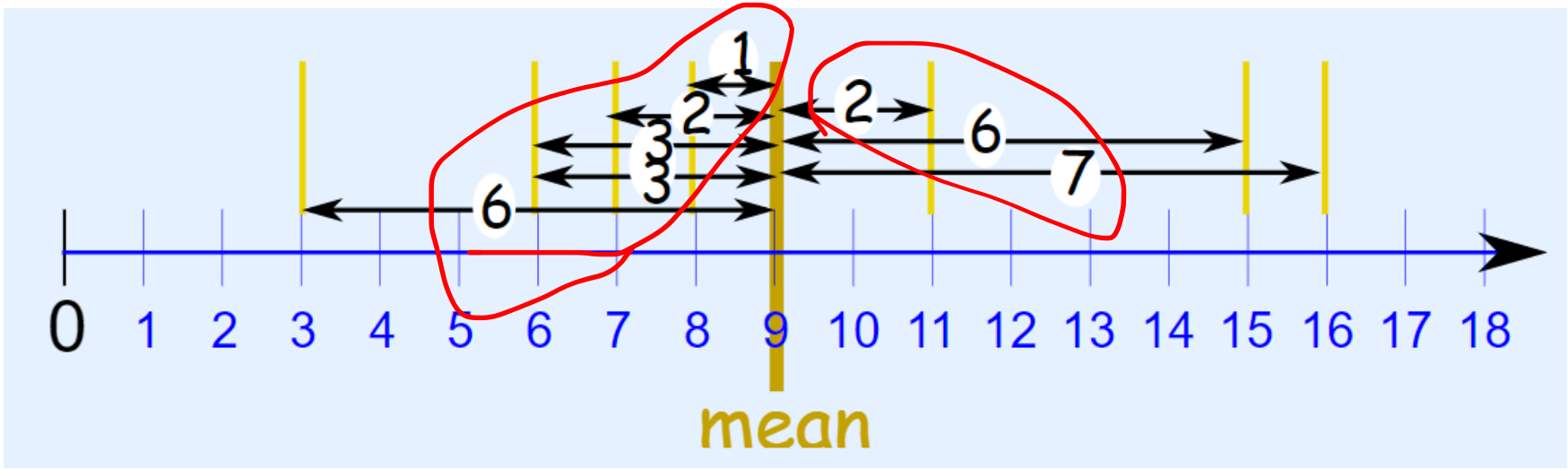
- Step 3. Find the **Mean Deviation**:

$$\text{Mean Deviation} = \frac{\Sigma|x - \mu|}{N} = \frac{636}{5} = 127.2$$

- So, on average, the dogs' heights are **127.2 mm from the mean**.
- (Compare that with the Standard Deviation of **147 mm**)

Mean Deviation: A Useful Check

- The deviations on **one side** of the mean should equal the deviations on the **other side**.
- From our first example:
- Example: 3, 6, 6, 7, 8, 11, 15, 16
- The deviations are:



$$6 + 3 + 3 + 2 + 1 = 2 + 6 + 7$$

$$15 = 15$$

Mean Deviation: A Useful Check

- Example: Dogs
- Deviations left of mean: 224 + 94 = 318

- Deviations right of mean: 206 + 76 + 36 = 318

- If they are not equal ... you may have made a msitake!

Standard Deviation

- The Standard Deviation is a measure of how spread out numbers are.
- Its symbol is **σ** (the greek letter sigma)
- The formula is easy: it is the **square root** of the **Variance.** So now you ask, "What is the Variance?"

Variance

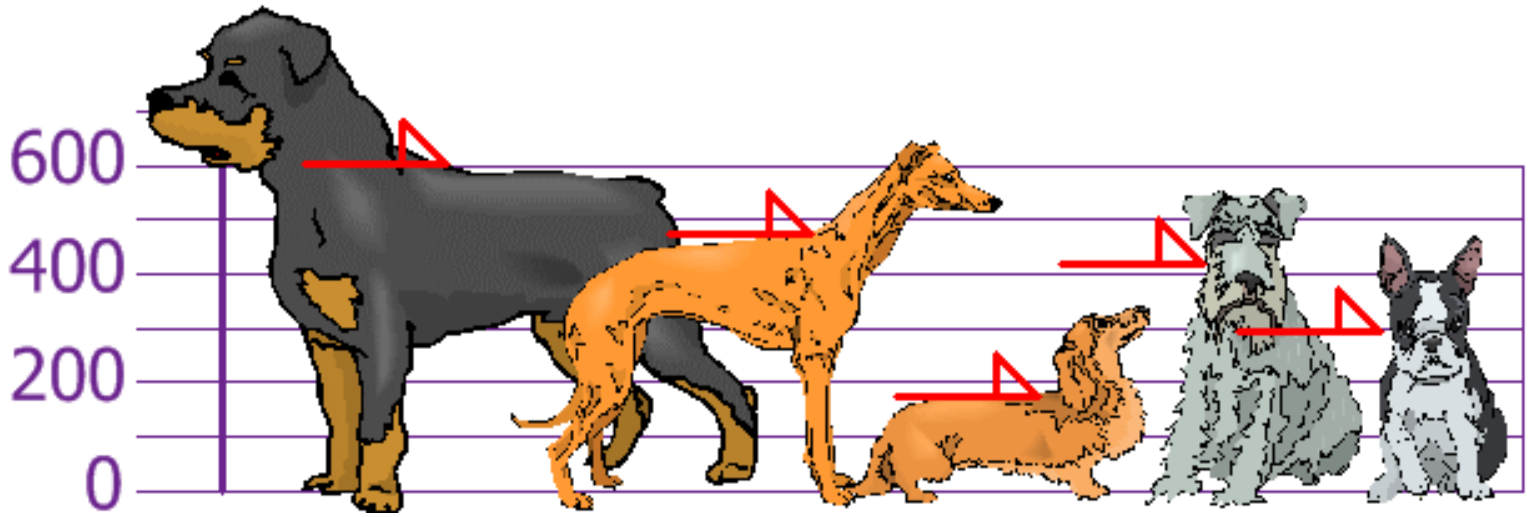
- The Variance is defined as:

  The average of the **squared** differences from the Mean.

- To calculate the variance follow these steps:
  - Work out the Mean
  - Then for each number: subtract the Mean and square the result
  - Then work out the average of those squared differences.

Example

- You and your friends have just measured the heights of your dogs (in millimeters):



- The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.
- Find out the Mean, the Variance, and the Standard Deviation.

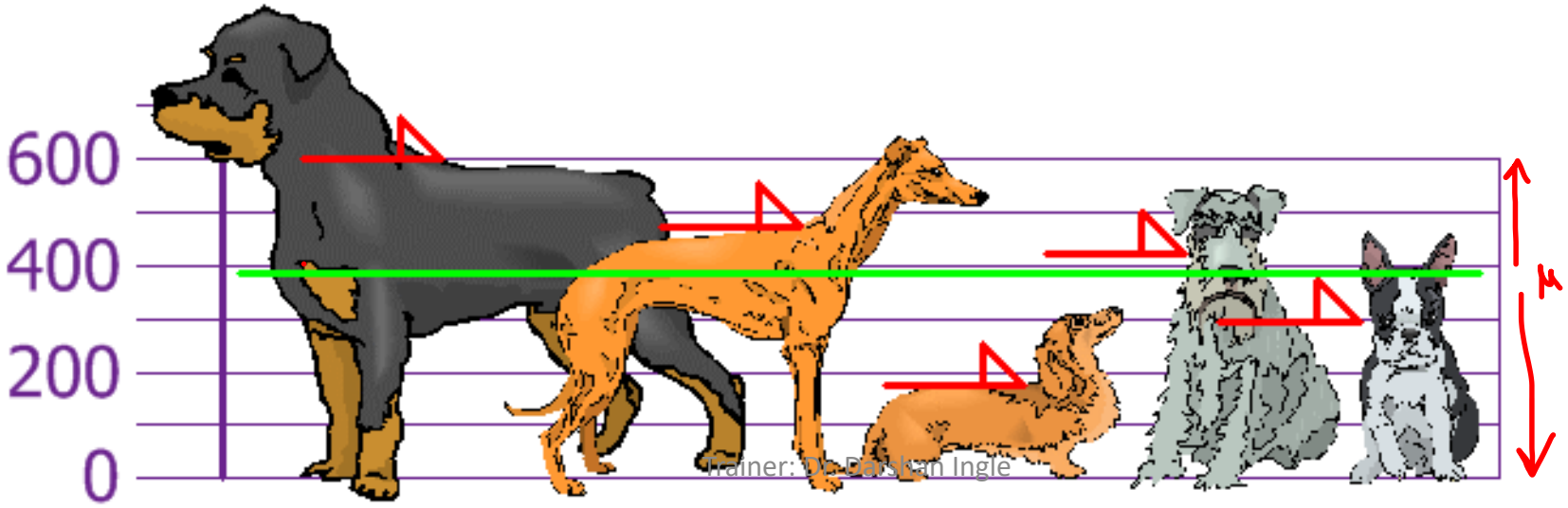
Example

$$\text{Mean} = \frac{600 + 470 + 170 + 430 + 300}{5}$$

$$= \frac{1970}{5}$$

$$= 394$$

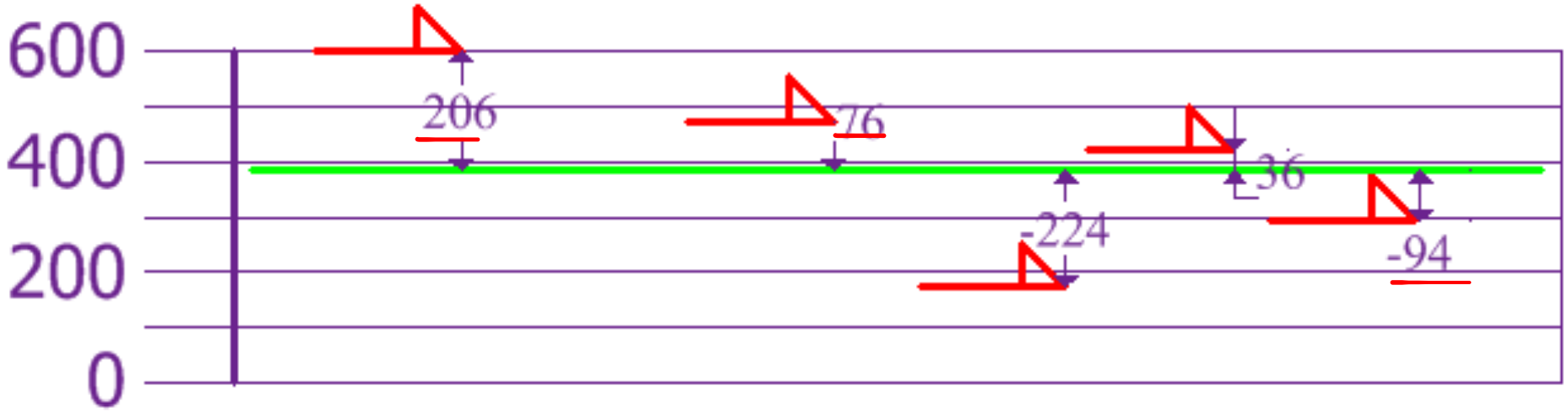
- so the mean (average) height is 394 mm. Let's plot this on the chart:



Trainer: Dr Rashan Ingle

Example

- Now we calculate each dog's difference from the Mean:



- To calculate the Variance, take each difference, square it, and then average the result:

Example

Variance

$$\sigma^2 = \frac{206^2 + 76^2 + (-224)^2 + 36^2 + (-94)^2}{5}$$

$$= \frac{42436 + 5776 + 50176 + 1296 + 8836}{5}$$

$$= \frac{108520}{5}$$

$$= 21704$$

- So the Variance is **21,704**

Example

- And the Standard Deviation is just the square root of Variance, so:
- Standard Deviation

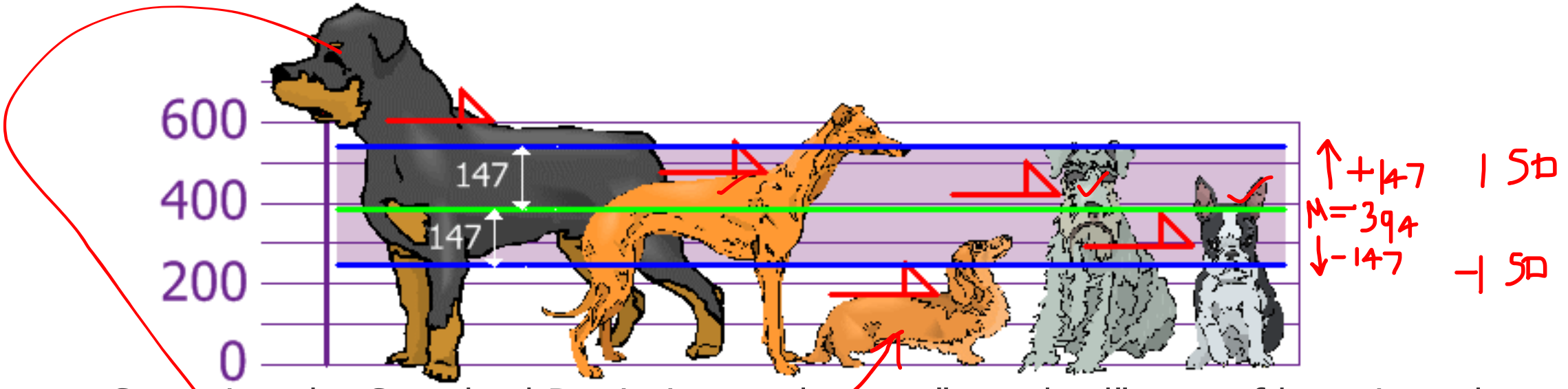
$$\sigma = \sqrt{21704}$$

$$= 147.32...$$

$$= \mathbf{147}\ \textit{(to the nearest mm)}$$

Example

- And the good thing about the Standard Deviation is that it is useful. Now we can show which heights are within one Standard Deviation (147mm) of the Mean:



- So, using the Standard Deviation we have a "standard" way of knowing what is normal, and what is extra large or extra small.
- Rottweilers **are** tall dogs. And Dachshunds **are** a bit short, right?

Standard Deviation of Sample data

- Our example has been for a **Population** (the 5 dogs are the only dogs we are interested in).
- But if the data is a **Sample** (a selection taken from a bigger Population), then the calculation changes!
- When you have "N" data values that are:
  - **The Population**: divide by **N** when calculating Variance (like we did)
  - **A Sample**: divide by **N-1** when calculating Variance

Standard Deviation of Sample data

- Example: if our 5 dogs are just a **sample** of a bigger population of dogs, we divide by **4 instead of 5** like this:
- Sample Variance = 108,520 / **4** = **27,130**
- Sample Standard Deviation = √27,130 = **165** (to the nearest mm)
- Think of it as a "correction" when your data is only a sample.

Formulaes

The "**Population** Standard Deviation":

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

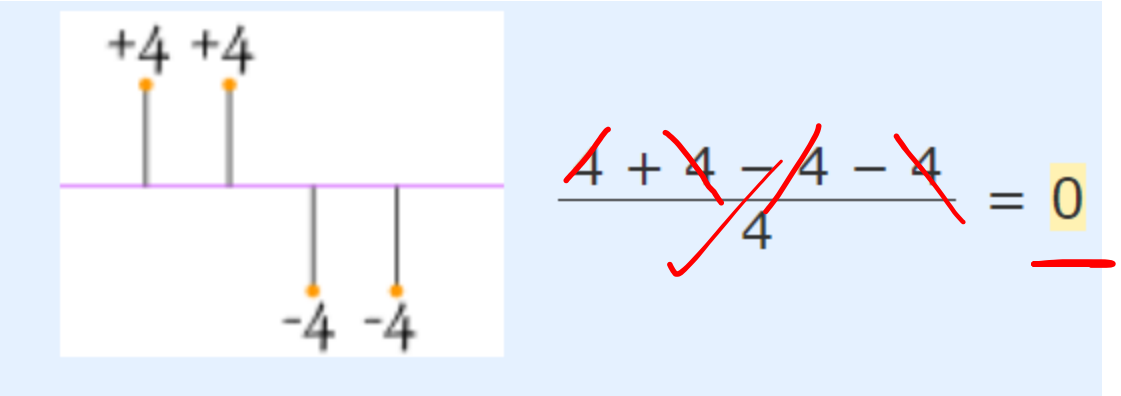
The "**Sample** Standard Deviation":

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2}$$

- Looks complicated, but the important change is to divide by **N-1** (instead of **N**) when calculating a Sample Variance.

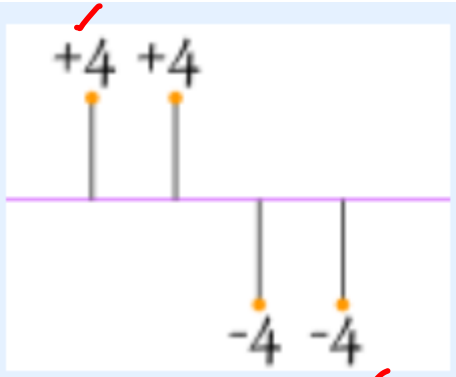
Why *square* the differences?

- If we just add up the differences from the mean ... the negatives cancel the positives:



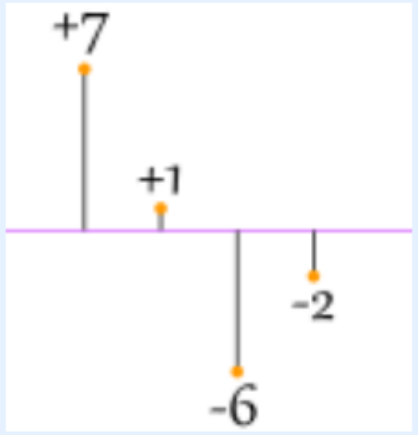
Why *square* the differences?

- So that won't work. How about we use absolute values?



$$\frac{|4| + |4| + |-4| + |-4|}{4} = \frac{4 + 4 + 4 + 4}{4} = 4$$

- That looks good (and is the Mean Deviation), but what about this case:

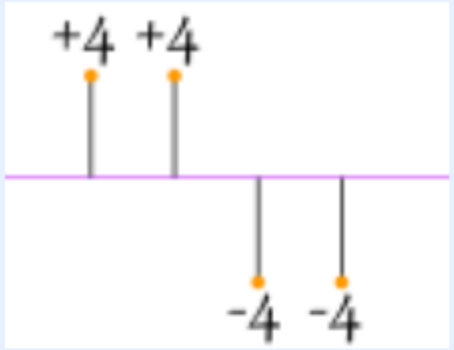


$$\frac{|7| + |1| + |-6| + |-2|}{4} = \frac{7 + 1 + 6 + 2}{4} = 4$$

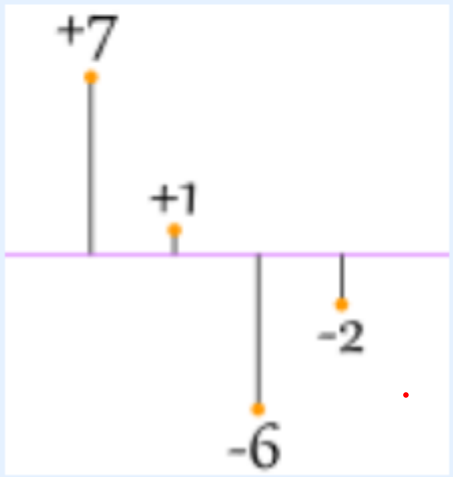
- Oh No! It also gives a value of 4, Even though the differences are more spread out.

Why *square* the differences?

- So let us try squaring each difference (and taking the square root at the end):



$$\sqrt{\left(\frac{4^2 + 4^2 + 4^2 + 4^2}{4}\right)} = \sqrt{\left(\frac{64}{4}\right)} = 4$$

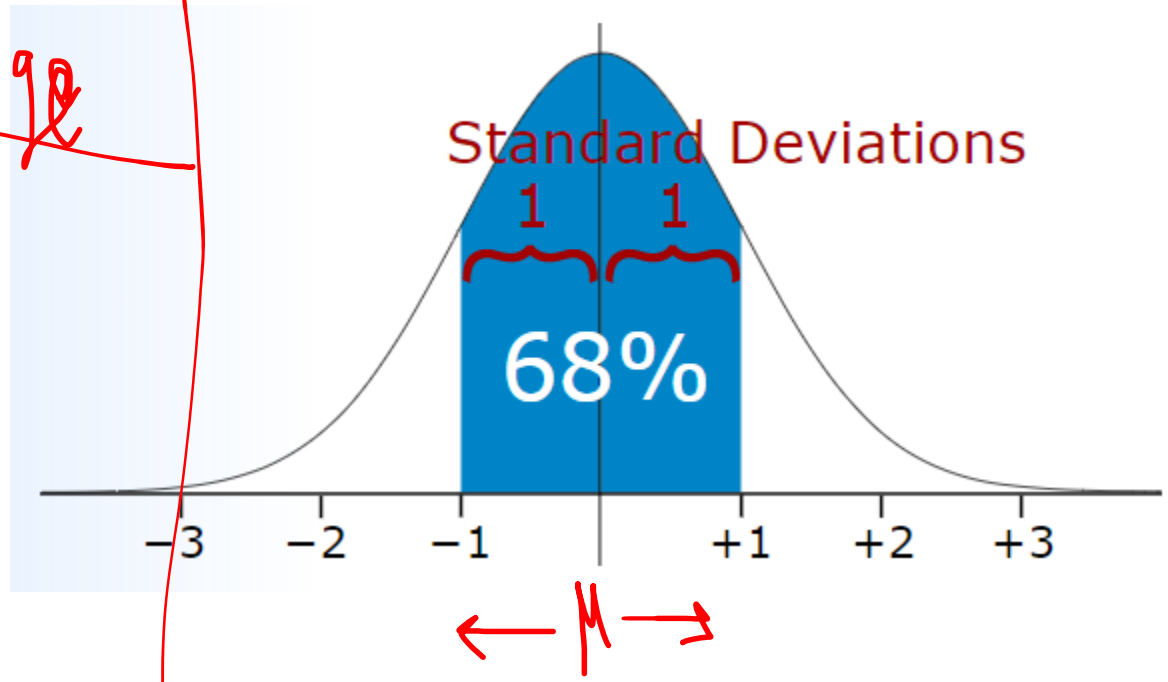


$$\sqrt{\left(\frac{7^2 + 1^2 + 6^2 + 2^2}{4}\right)} = \sqrt{\left(\frac{90}{4}\right)} = 4.74\ldots$$

- That is nice! The Standard Deviation is bigger when the differences are more spread out ... just what we want.

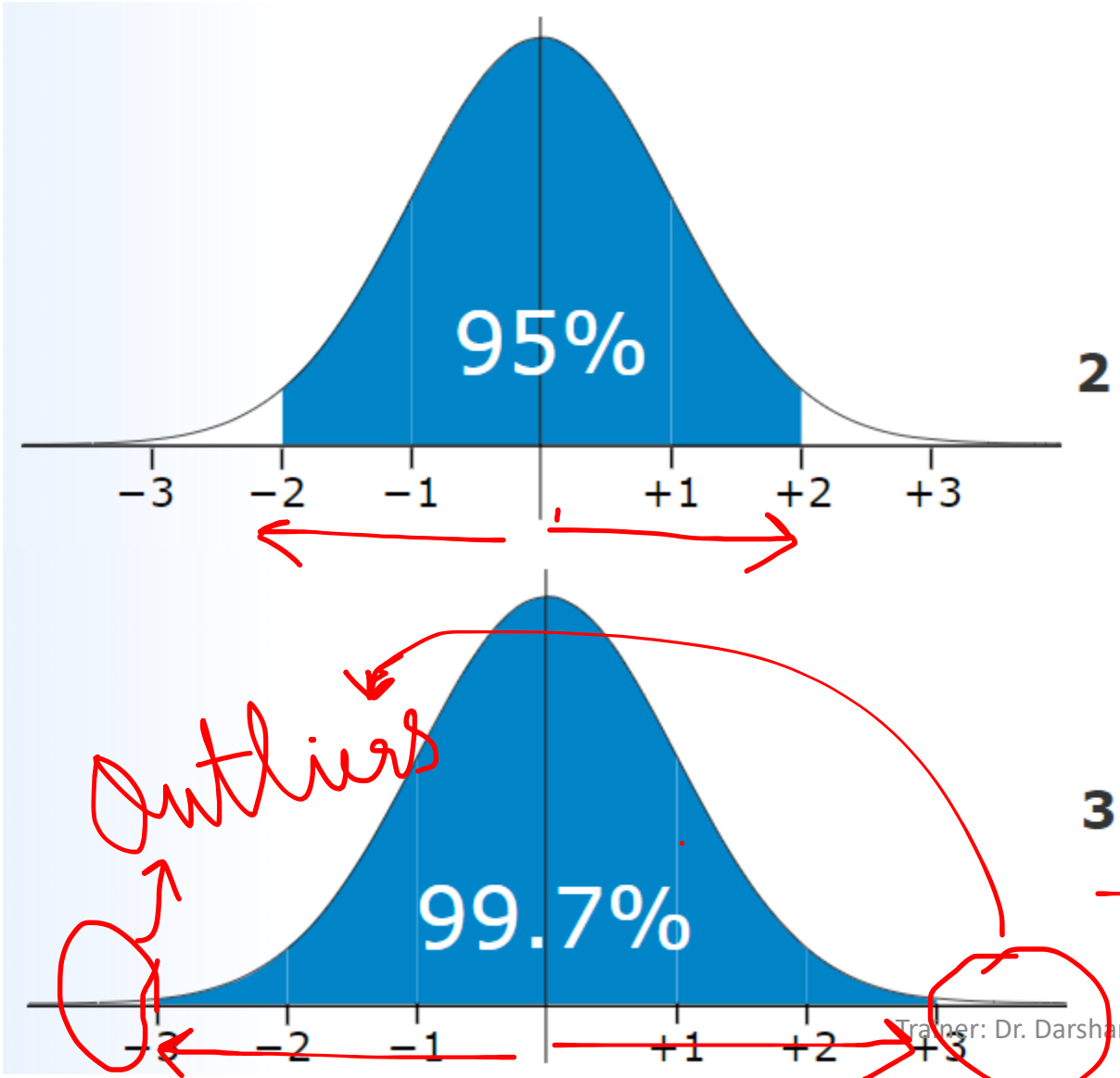
Range of Standard Deviation

- We can expect about 68% of values to be within plus-or-minus 1 standard deviation.



68% of values are within
1 standard deviation of the mean

Range of Standard Deviation

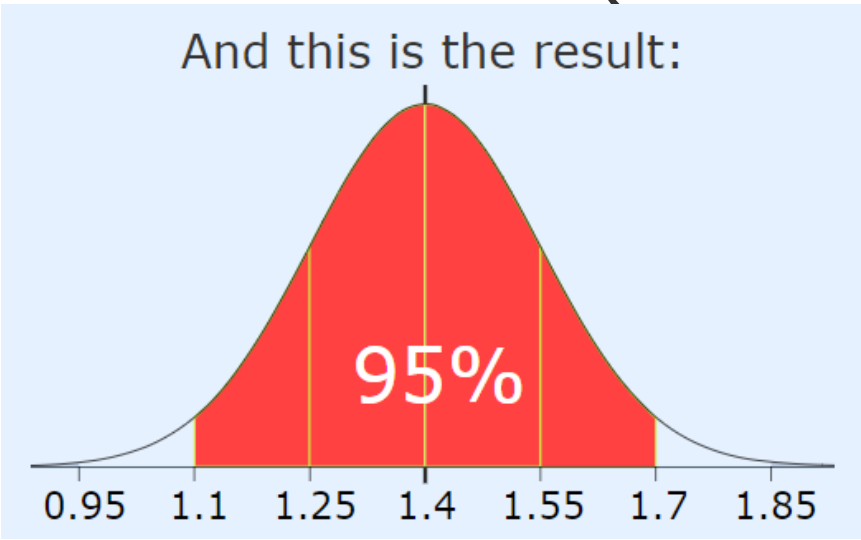


95% of values are within
2 standard deviations of the mean

99.7% of values are within
3 standard deviations of the mean

Standard Deviation

- Example: 95% of students at school are between **1.1m and 1.7m** tall.
- Assuming this data is **normally distributed** can you calculate the mean and standard deviation?
- The mean is halfway between 1.1m and 1.7m:
  - Mean = (1.1m + 1.7m) / 2 = **1.4m**
- 95% is 2 standard deviations either side of the mean (a total of 4 standard deviations) so:
  - 1 standard deviation = (1.7m-1.1m) / 4
    = 0.6m / 4
    = **0.15m**



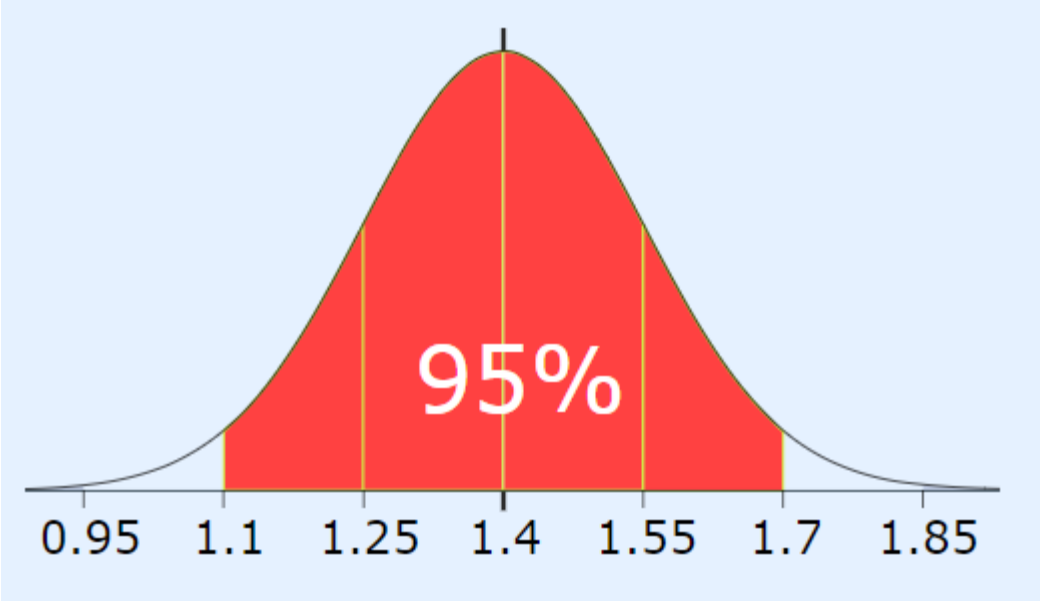
Range of Standard Deviation

- It is good to know the standard deviation, because we can say that any value is:
- **likely** to be within 1 standard deviation (68 out of 100 should be)
- **very likely** to be within 2 standard deviations (95 out of 100 should be)
- **almost certainly** within 3 standard deviations (997 out of 1000 should be)

Standard Scores

- The number of **standard deviations from the mean** is also called the "Standard Score", "sigma" or "z-score". Get used to those words!
- Example: In that same school one of your friends is 1.85m tall

  - You can see on the bell curve that 1.85m is **3 standard deviations** from the mean of 1.4, so:
  - Your friend's height has a "z-score" of 3.0



Standard Scores

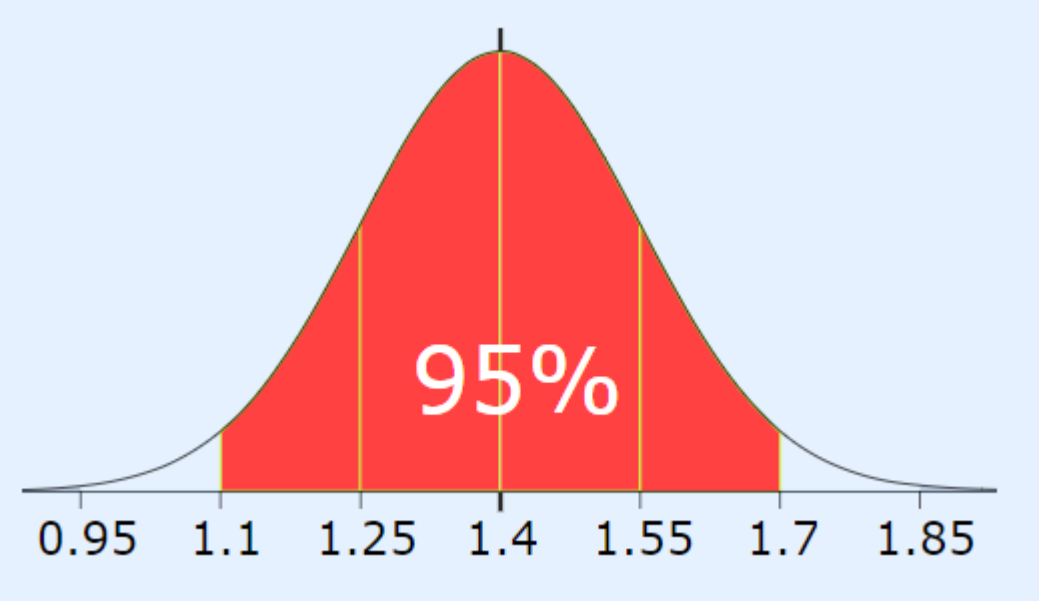
It is also possible to **calculate** how many standard deviations 1.85 is from the mean

How far is 1.85 from the mean?

It is 1.85 - 1.4 = **0.45m from the mean**

How many standard deviations is that? The standard deviation is 0.15m, so:

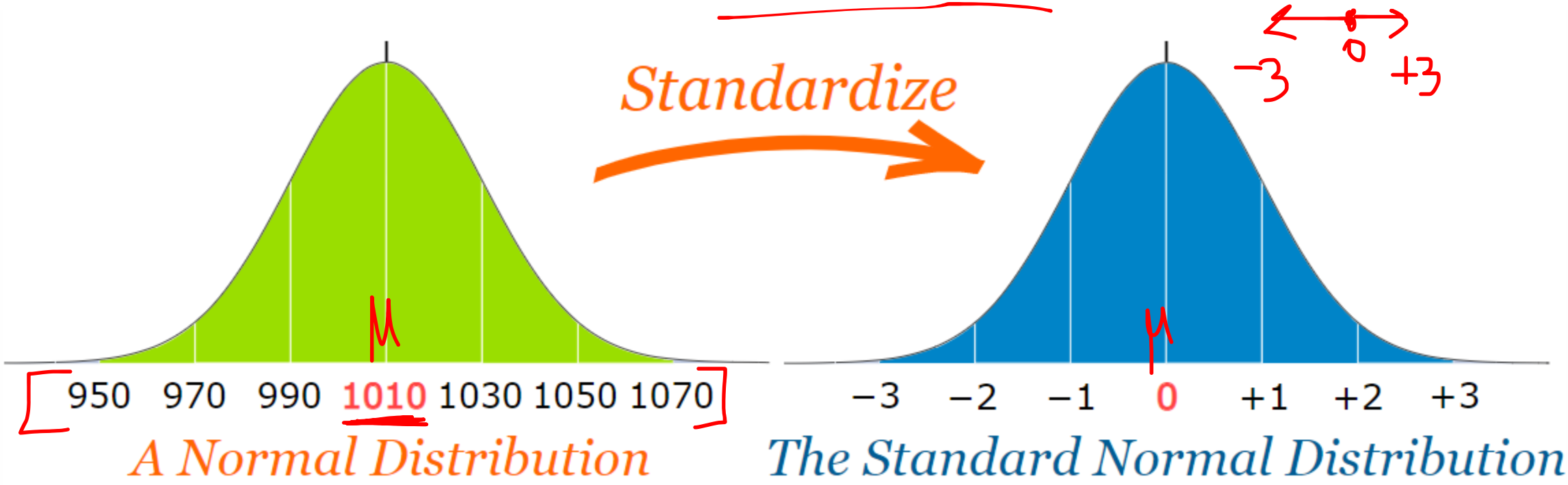
0.45m / 0.15m = **3 standard deviations**



Standard Scores

$$Z = \frac{x - \mu}{\sigma}$$

- So to convert a value to a Standard Score ("z-score"):
  - first subtract the mean,
  - then divide by the Standard Deviation
- And doing that is called "Standardizing":



Standard Scores

- We can take any Normal Distribution and convert it to The Standard Normal Distribution.
- Example: Travel Time
- A survey of daily travel time had these results (in minutes):
- 26, 33, 65, 28, 34, 55, 25, 44, 50, 36, 26, 37, 43, 62, 35, 38, 45, 32, 28, 34
- The **Mean is 38.8 minutes**, and the **Standard Deviation is 11.4 minutes**
- Convert the values to z-scores ("standard scores").
-
- To convert **26**:
- first subtract the mean: 26 − 38.8 = −12.8,
- then divide by the Standard Deviation: −12.8/11.4 = −1.12
- So **26** is **−1.12 Standard Deviations** from the Mean
-

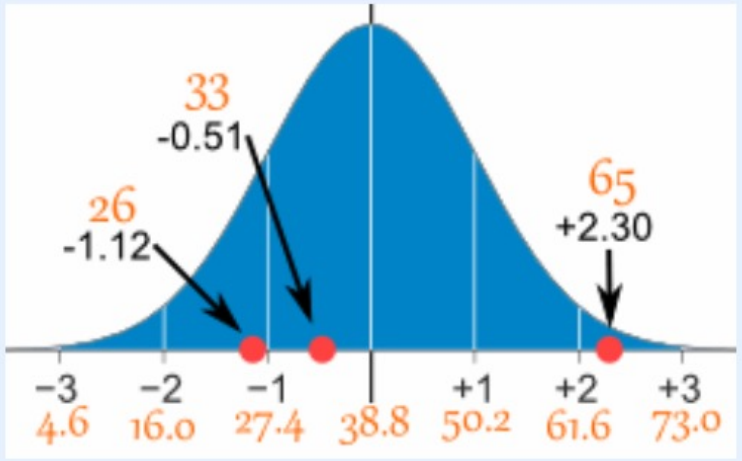
$$\frac{26 - 38.8}{11.4}$$

Standard Scores

- Here are the first three conversions

Original Value	Calculation	Standard Score (z-score)
26	(26-38.8) / 11.4 =	**−1.12**
33	(33-38.8) / 11.4 =	**−0.51**
65	(65-38.8) / 11.4 =	**+2.30**
...	...	...

And here they are graphically:



Z-Score formula

$$z = \frac{x - \mu}{\sigma}$$

- **z** is the "z-score" (Standard Score)
- **x** is the value to be standardized
- **μ** ('mu") is the mean
- **σ** ("sigma") is the standard deviation

Example: Travel Time (continued)

- $\mu = 38.8$
- $\sigma = 11.4$
- Here are the first three conversions using the "z-score formula":

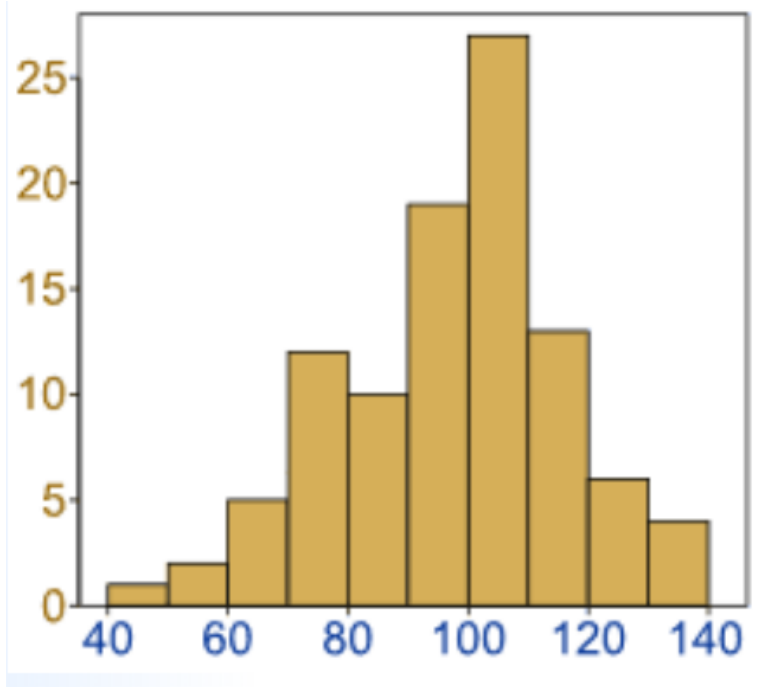
x	$\frac{x - \mu}{\sigma}$	z (z-score)
26	$\frac{26 - 38.8}{11.4}$	$= -1.12$
33	$\frac{33 - 38.8}{11.4}$	$= -0.51$
65	$\frac{65 - 38.8}{11.4}$	$= +2.30$
...	...	...

Why Standardize ... ?

- It can help us make decisions about our data.
- Example: Professor Willoughby is marking a test.
- Here are the student's results (out of 60 points):
  - 20, 15, 26, 32, 18, 28, 35, 14, 26, 22, 17
- Most students didn't even get 30 out of 60, and **most will fail**.
- The test must have been really hard, so the Prof decides to Standardize all the scores and only fail people 1 standard deviation below the mean.
- The **Mean is 23**, and the **Standard Deviation is 6.6**, and these are the Standard Scores:
  - -0.45, -1.21, 0.45, 1.36, -0.76, 0.76, 1.82, -1.36, 0.45, -0.15, -0.91
- Now only 2 students will fail (the ones lower than −1 standard deviation)
- Much fairer!

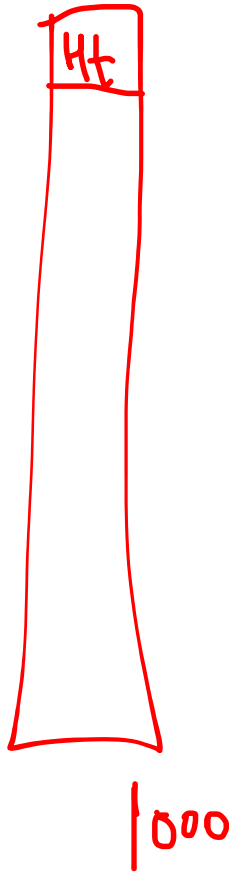
Histograms

- **Histogram**: a graphical display of data using bars of different heights.
- It is similar to a Bar Chart, but a histogram groups numbers into **ranges** .
- The height of each bar shows how many fall into each range.
- And you decide what ranges to use!



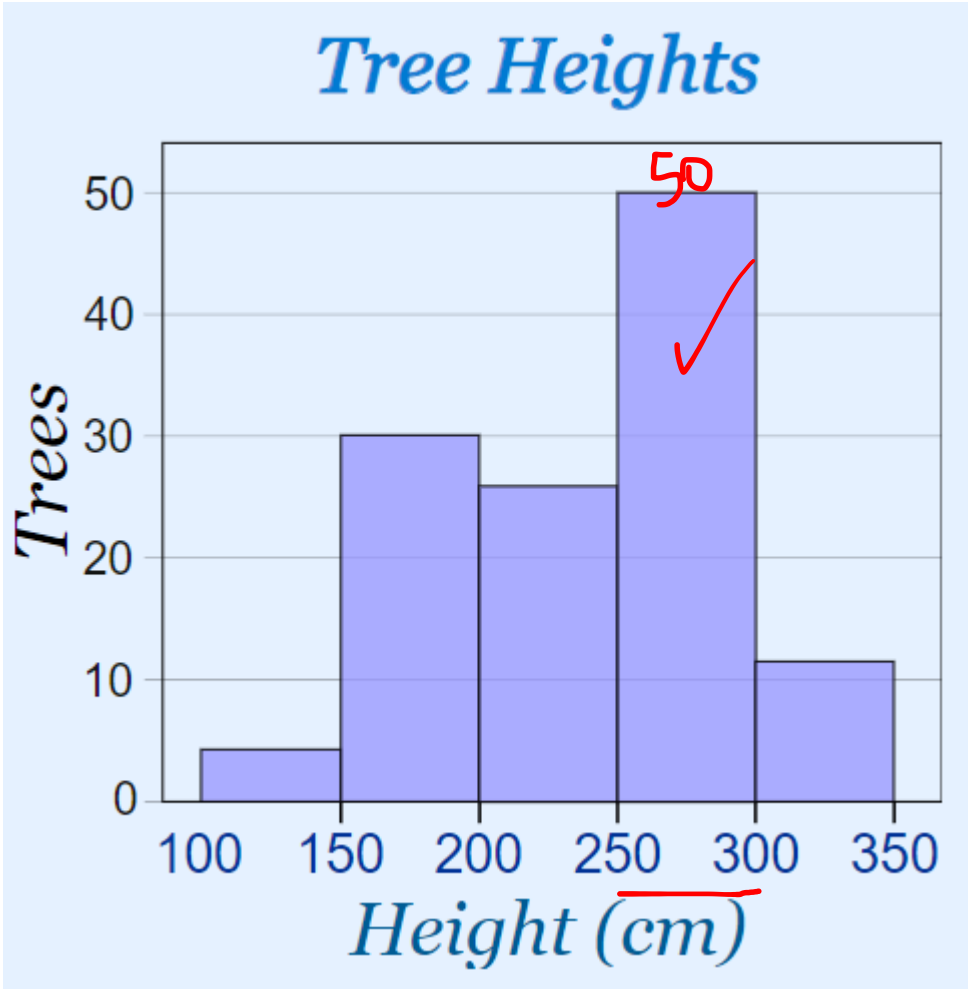
Histograms

- Example: Height of Orange Trees
- You measure the height of every tree in the orchard in centimeters (cm)
- The heights vary from 100 cm to 340 cm
- You decide to put the results into groups of 50 cm:
- The **100 to just below 150 cm** range,
- The **150 to just below 200 cm** range,
- etc...
- So a tree that is 260 cm tall is added to the "250-300" range.



Histograms

- And here is the result:



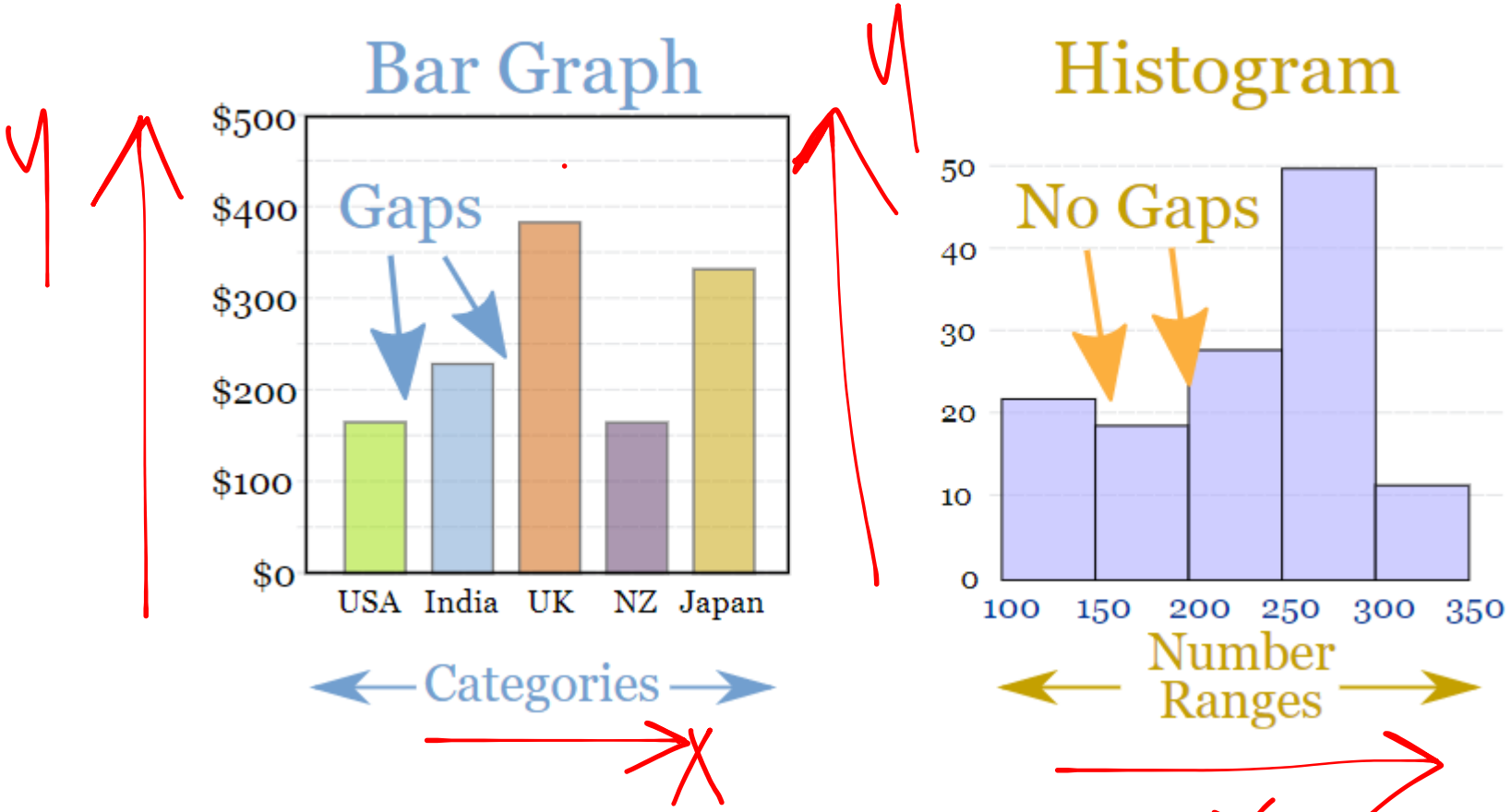
- You can see (for example) that there are **30** trees from **150 cm to just below 200 cm tall**

Histograms

- Histograms are a great way to show results of continuous data, such as:
- weight
- height
- how much time
- etc.

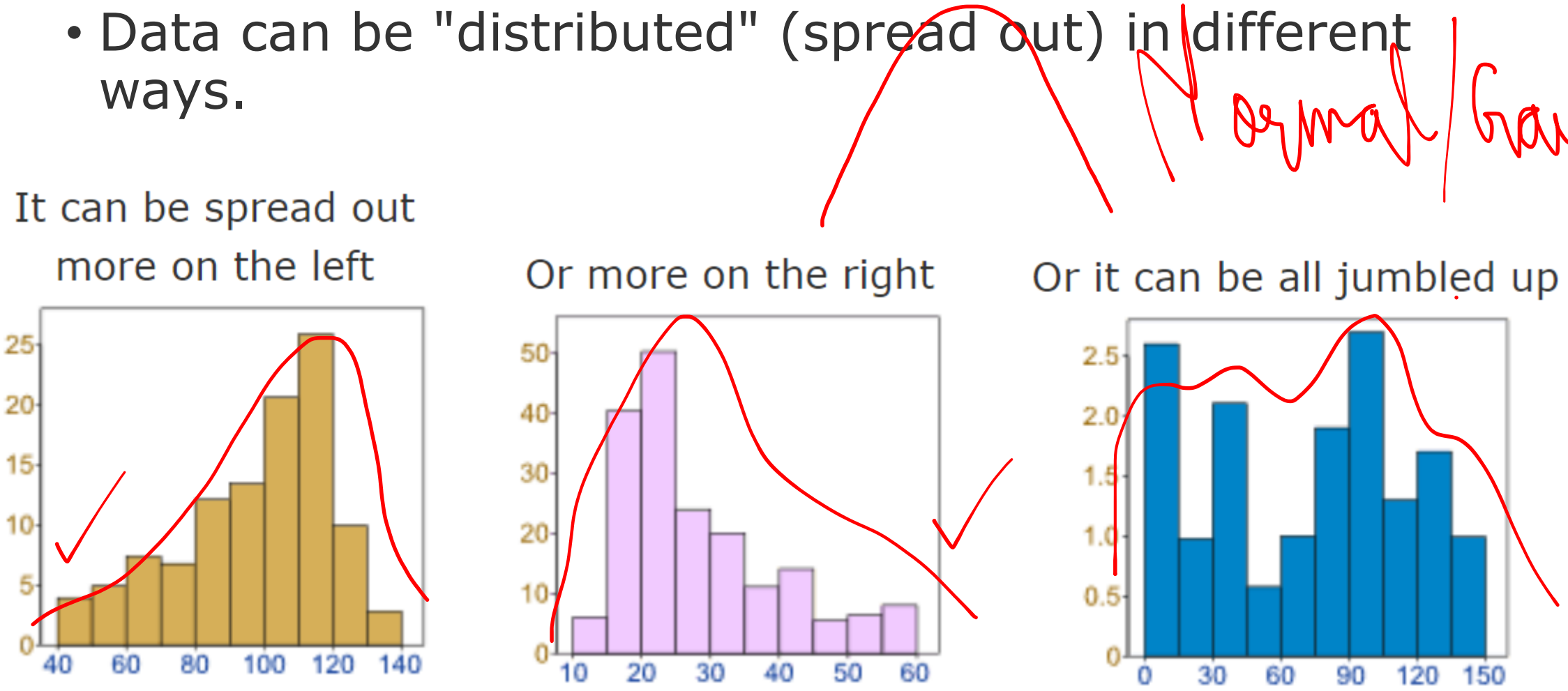
Histograms

- But when the data is in **categories** (such as Country or Favorite Movie), we should use a Bar Chart.



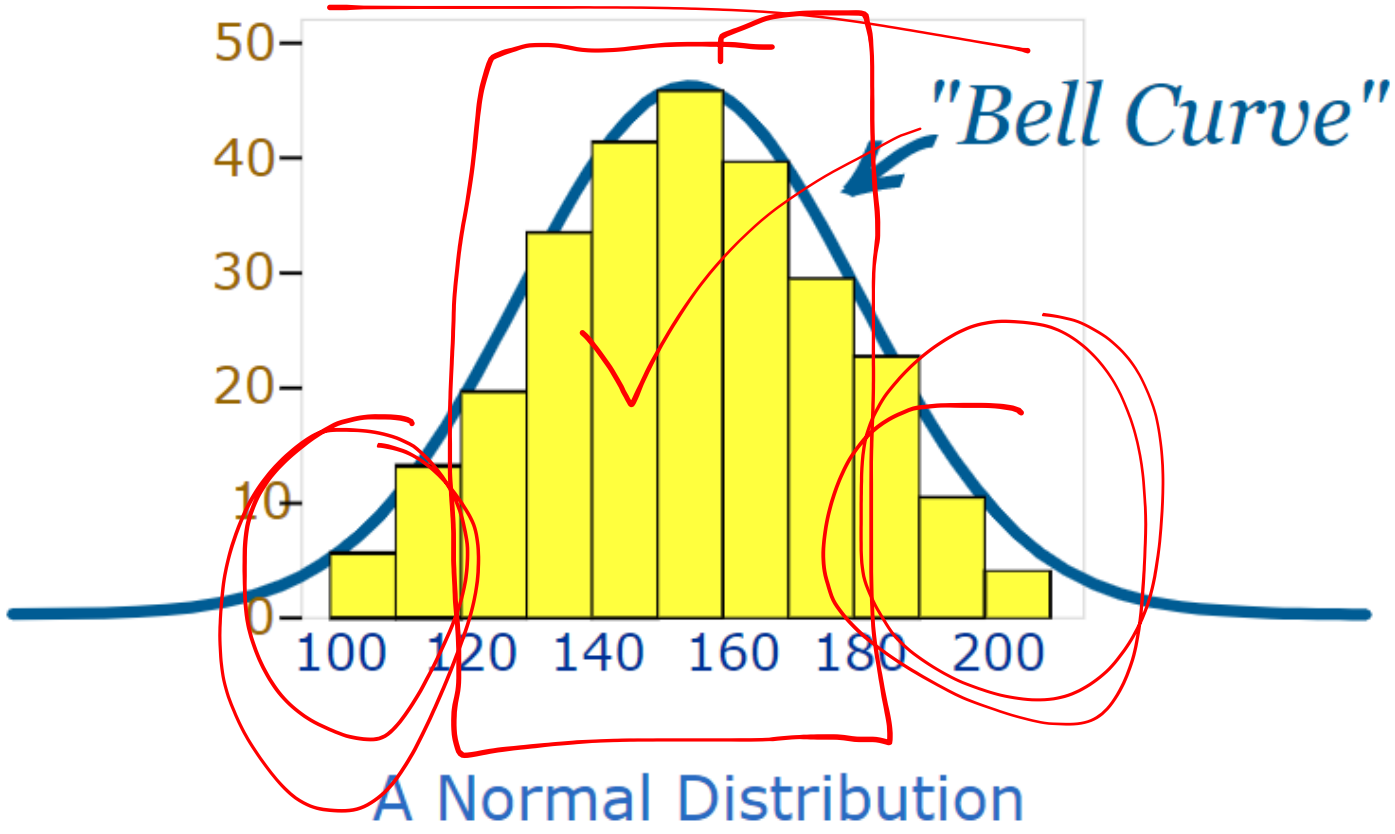
Normal Distribution

- Data can be "distributed" (spread out) in different ways.

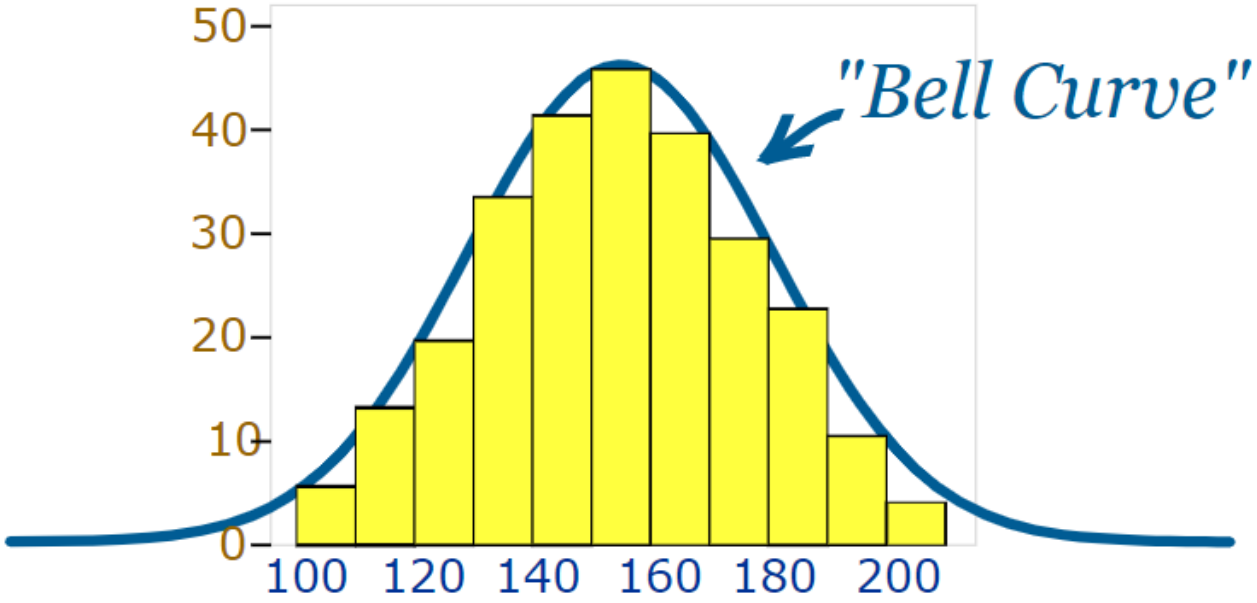


Normal Distribution

- But there are many cases where the data tends to be around a central value with no bias left or right, and it gets close to a "Normal Distribution" like this:



Normal Distribution



A Normal Distribution

- The "Bell Curve" is a Normal Distribution. And the yellow histogram shows some data that follows it closely, but not perfectly (which is usual).

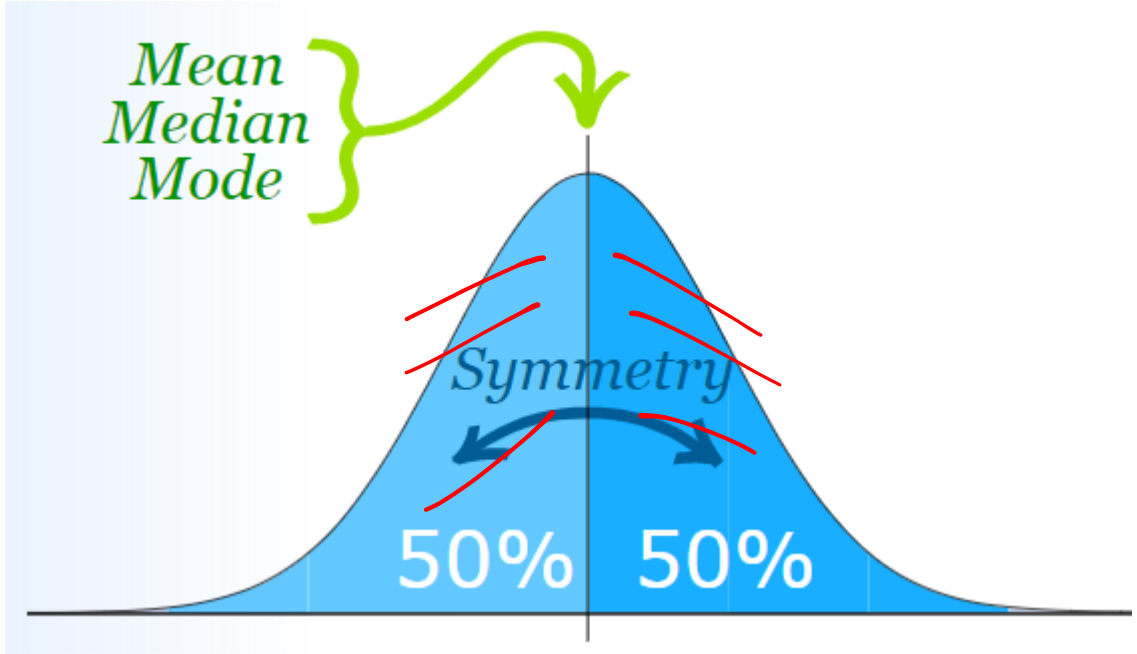
Normal Distribution

- It is often called a "Bell Curve‟ because it looks like a bell.
- Many things closely follow a Normal Distribution:

1. heights of people
2. size of things produced by machines
3. errors in measurements
4. blood pressure
5. marks on a test

Normal Distribution

- We say the data is "normally distributed":



The **Normal Distribution** has:

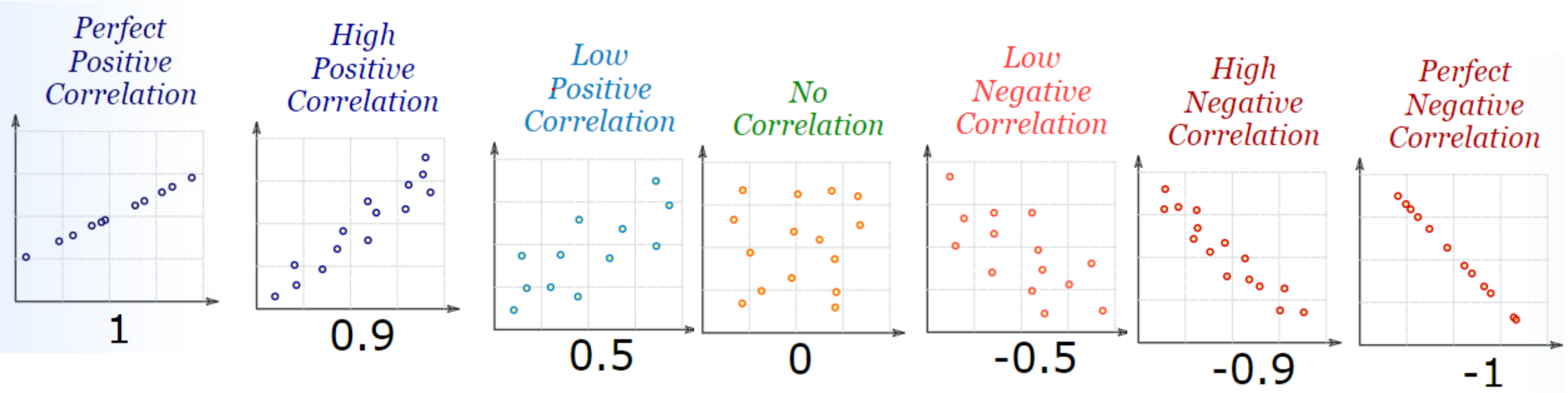
- mean = median = mode
- symmetry about the center
- 50% of values less than the mean and 50% greater than the mean

Normal Distribution

- **Quincunx**
- You can see a normal distribution being created by random chance!
- It is called the Quincunx and it is an amazing machine.
- Have a play with it!
- https://www.youtube.com/watch?v=3m4bxse2JEQ
- https://www.transum.org/Maths/Activity/PhET/Plinko_Probability.asp

Correlation

- When two sets of data are strongly linked together we say they have a **High Correlation**.
- The word Correlation is made of **Co-** (meaning "together"), and **Relation**
- Correlation is **Positive** when the values **increase** together, and
- Correlation is **Negative** when one value **decreases** as the other increases
- A correlation is assumed to be **linear** (following a line).



Correlation

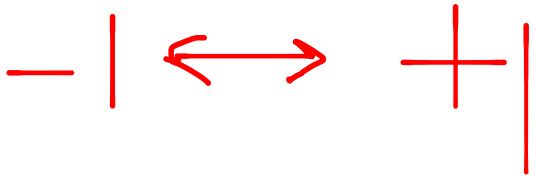
- Correlation can have a value:
  - 1 is a perfect positive correlation
  - 0 is no correlation (the values don't seem linked at all)
  - -1 is a perfect negative correlation
- The value shows **how good the correlation is** (not how steep the line is), and if it is positive or negative.
- **Correlation Is Not Good at Curves**
- **Ex:** ice cream shop keeps track of how much ice cream they sell versus the temperature on that day
- The correlation calculation only works properly for straight line relationships.
- Ex: **there has been a heat wave!** It gets so hot that people aren't going near the shop, and **sales start dropping**.

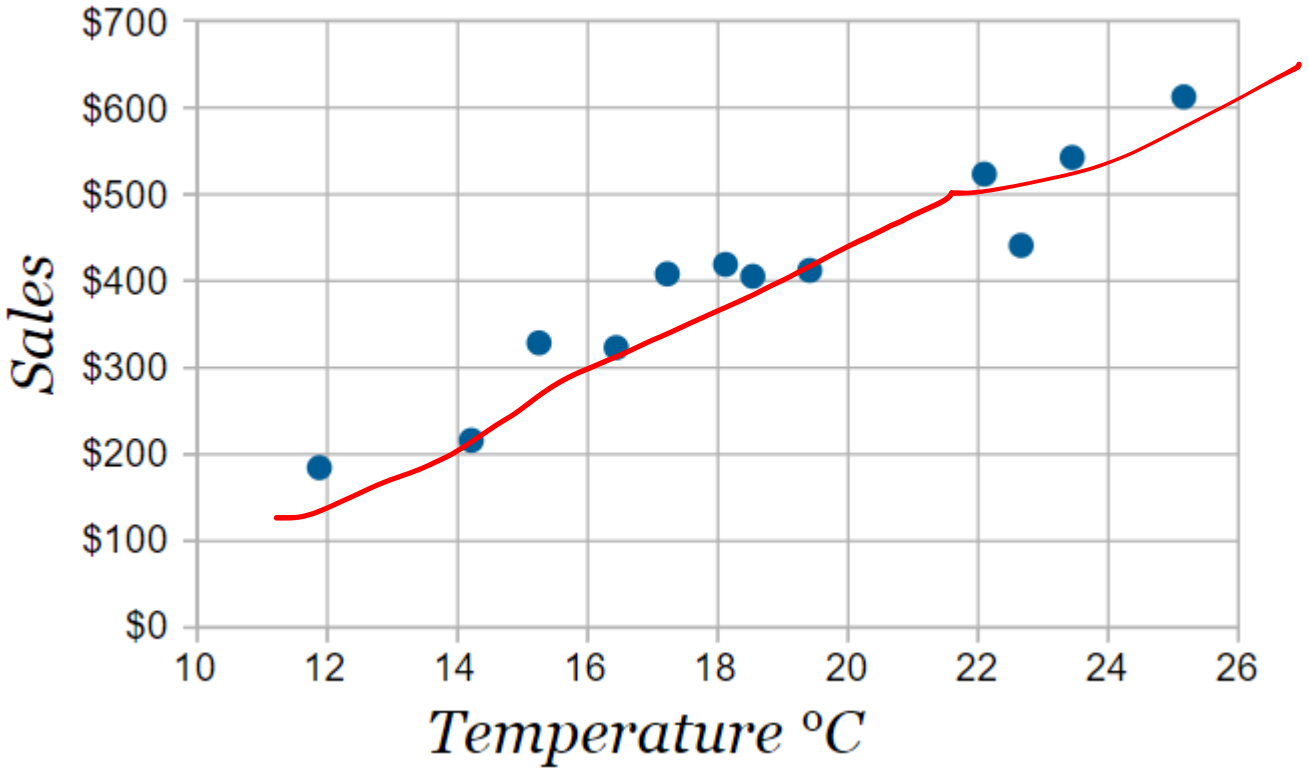
Pearsons Corelation Coefficient

- **Example: Ice Cream Sales**
- The local ice cream shop keeps track of how much ice cream they sell versus the temperature on that day, here are their figures for the last 12 days:

Ice Cream Sales vs Temperature	
Temperature °C	**Ice Cream Sales**
14.2°	$215
16.4°	$325
11.9°	$185
15.2°	$332
18.5°	$406
22.1°	$522
19.4°	$412
25.1°	$614
23.4°	$544
18.1°	$421
22.6°	$445
17.2°	$408

Pearsons Corelation Coefficient





- We can easily see that warmer weather and higher sales go together. The relationship is good but not perfect.
- In fact the Pearson's correlation is **0.9575**

Pearsons Corelation Coefficient

- The correlation coefficient ranges from −1 to 1.

✓A value of 1 implies that a linear equation describes the relationship between *X* and *Y* perfectly, with all data points lying on a line for which *Y* increases as *X* increases.

✓A value of −1 implies that all data points lie on a line for which *Y* decreases as *X* increases.

✓A value of 0 implies that there is no linear correlation between the variables.

Pearsons Corelation Coefficient Formula

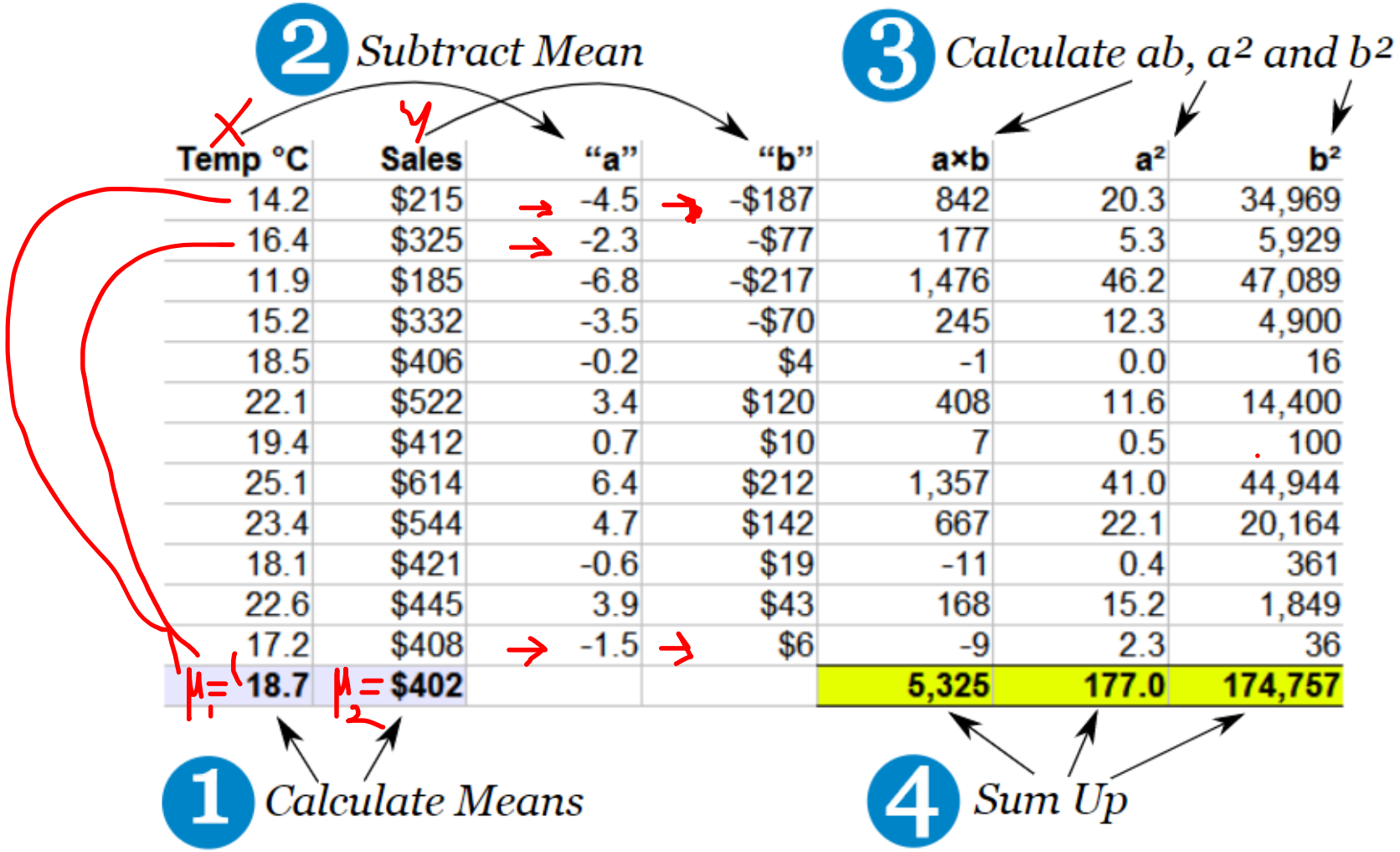
$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

- Where:
- Σ is Sigma, the symbol for "sum up"
- correlation formula (xi - xbar) is each x-value minus the mean of x (called "a" in the next slide)
- correlation formula (yi - ybar) is each y-value minus the mean of y (called "b" in the next slide)

Calculating Pearsons Corelation Coefficient

- Let us call the two sets of data "x" and "y" (in our case Temperature is **x** and Ice Cream Sales is **y**):
- Step 1: Find the mean of **x**, and the mean of **y**
- Step 2: Subtract the mean of x from every x value (call them "**a**"), and subtract the mean of y from every y value (call them "**b**")
- Step 3: Calculate: **ab**, $\mathbf{a^2}$ and $\mathbf{b^2}$ for every value
- Step 4: Sum up **ab**, sum up $\mathbf{a^2}$ and sum up $\mathbf{b^2}$
- Step 5: Divide the sum of ab by the square root of [(sum of a^2) × (sum of b^2)]

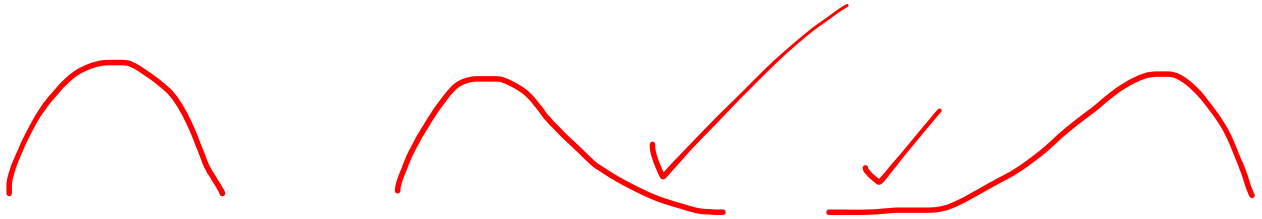
Calculating Pearsons Corelation Coefficient



Temp °C	Sales	"a"	"b"	a×b	a²	b²
14.2	$215	-4.5	-$187	842	20.3	34,969
16.4	$325	-2.3	-$77	177	5.3	5,929
11.9	$185	-6.8	-$217	1,476	46.2	47,089
15.2	$332	-3.5	-$70	245	12.3	4,900
18.5	$406	-0.2	$4	-1	0.0	16
22.1	$522	3.4	$120	408	11.6	14,400
19.4	$412	0.7	$10	7	0.5	100
25.1	$614	6.4	$212	1,357	41.0	44,944
23.4	$544	4.7	$142	667	22.1	20,164
18.1	$421	-0.6	$19	-11	0.4	361
22.6	$445	3.9	$43	168	15.2	1,849
17.2	$408	-1.5	$6	-9	2.3	36
μ_1 = 18.7	μ_2 = $402			5,325	177.0	174,757

5 $$\frac{5{,}325}{\sqrt{177.0 \times 174{,}757}} = 0.9575$$

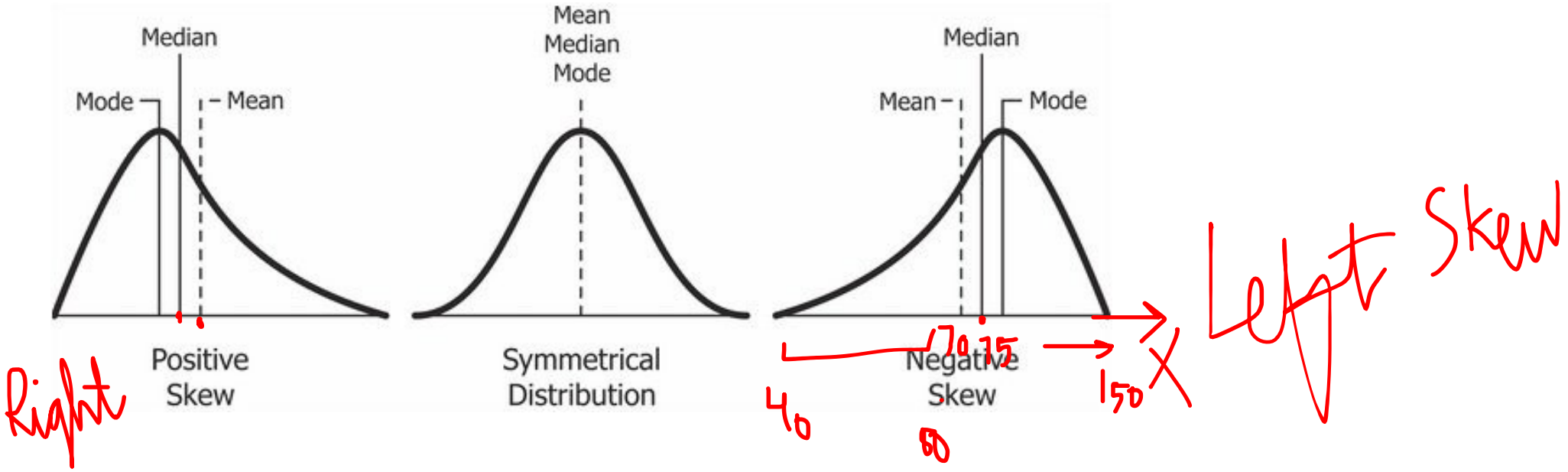
Skewness



- It is the *degree of distortion* from the symmetrical bell curve or the normal distribution.
- It measures the lack of symmetry in data distribution.
  It differentiates extreme values in one versus the other tail.
- A symmetrical distribution will have a skewness of 0.

Skewness

- There are two types of Skewness: Positive and Negative



- **Positive Skewness** means when the tail on the right side of the distribution is longer or fatter. The mean and median will be greater than the mode.
- **Negative Skewness** is when the tail of the left side of the distribution is longer or fatter than the tail on the right side. The mean and median will be less than the mode.

Skewness

$$g_1 = \frac{\sum_{i=1}^{N} (Y_i - \bar{Y})^3 / N}{s^3}$$

- **So, when is the skewness too much?**
- The rule of thumb seems to be:

✓If the skewness is between -0.5 and 0.5, the data are fairly symmetrical.

✓If the skewness is between -1 and -0.5(negatively skewed) or between 0.5 and 1(positively skewed), the data are moderately skewed.

✓If the skewness is less than -1(negatively skewed) or greater than 1(positively skewed), the data are highly skewed.

Skewness

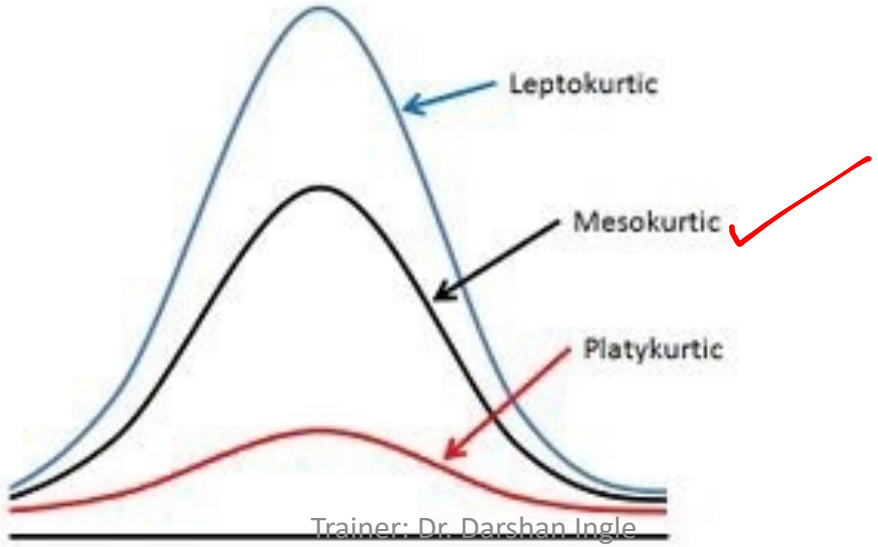
- **Example**
- Let us take a very common example of house prices. Suppose we have house values ranging from $100k to $1,000,000 with the average being $500,000.
- If the peak of the distribution was left of the average value, portraying a *positive skewness* in the distribution. It would mean that many houses were being sold for less than the average value, i.e. $500k. This could be for many reasons, but we are not going to interpret those reasons here.
- If the peak of the distributed data was right of the average value, that would mean a *negative skew.* This would mean that the houses were being sold for more than the average value.

Kurtosis

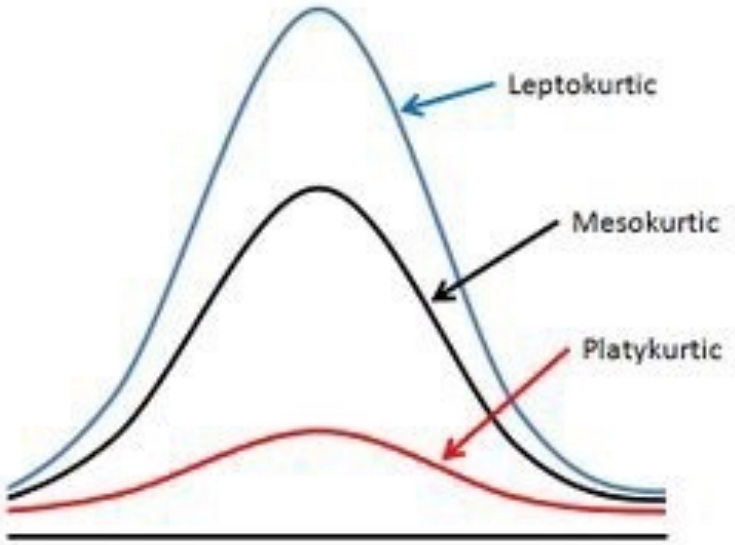
- Kurtosis is all about the tails of the distribution — not the peakedness or flatness.
- It is used to describe the extreme values in one versus the other tail.
- It is actually the *measure of outliers* present in the distribution.

Kurtosis

- **High kurtosis** in a data set is an indicator that data has heavy tails or outliers. If there is a high kurtosis, then, we need to investigate why do we have so many outliers. It indicates a lot of things, maybe wrong data entry or other things. Investigate!
  Low kurtosis in a data set is an indicator that data has light tails or lack of outliers. If we get low kurtosis(too good to be true), then also we need to investigate and trim the dataset of unwanted results.

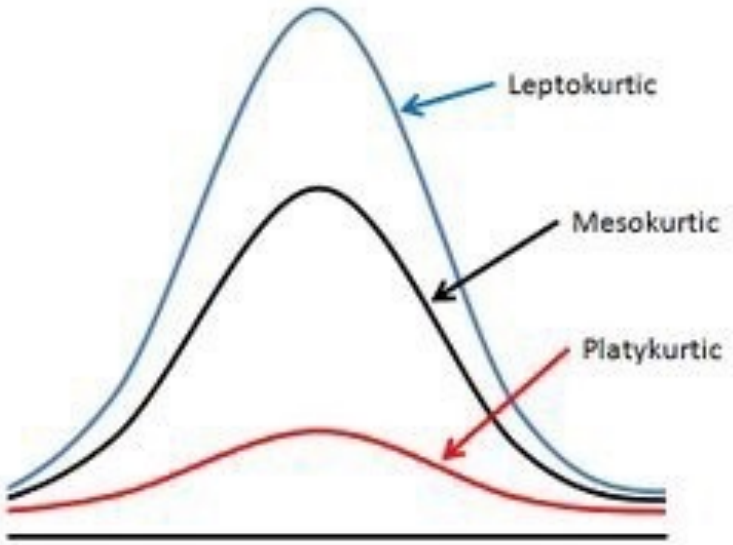


Kurtosis



- **Mesokurtic:** This distribution has kurtosis statistic similar to that of the normal distribution. It means that the extreme values of the distribution are similar to that of a normal distribution characteristic. This definition is used so that the standard normal distribution has a kurtosis of three.

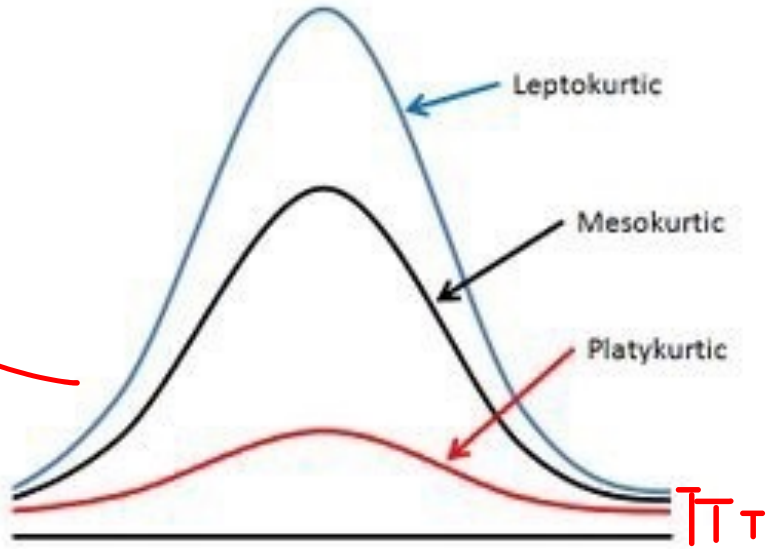
Kurtosis



- **Leptokurtic (Kurtosis > 3):** Distribution is longer, tails are fatter. Peak is higher and sharper than Mesokurtic, which means that data are heavy-tailed or profusion of outliers.
- Outliers stretch the horizontal axis of the histogram graph, which makes the bulk of the data appear in a narrow ("skinny") vertical range, thereby giving the "skinniness" of a leptokurtic distribution.

Kurtosis

$$\text{kurtosis} = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y})^4 / N}{s^4}$$



- **Platykurtic: (Kurtosis < 3):** Distribution is shorter, tails are thinner than the normal distribution. The peak is lower and broader than Mesokurtic, which means that data are light-tailed or lack of outliers.
- The reason for this is because the extreme values are less than that of the normal distribution.

Thank You